RAG Assignment: Comparing Retrieval and Re-Ranking Methods

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Code: `https://github.com/meet6868/RAG_Biencoder_CrossEncoder`

1 Dataset

Type: a single, complex **legal / regulatory PDF**, the EU **General Data Protection Regulation (GDPR)**, Regulation (EU) 2016/679, in its official 88-page version. It suits this assignment because it is full of exceptions ("unless", "shall not be required if"), prohibitions and passages that look alike.

Source	Official Journal of the EU (EUR-Lex), file `gdpr.pdf`
Size	88 pages, 173 recitals (explanations) + 99 articles (binding rules)
Cleaning	page headers and footnotes removed (footnotes found by their smaller font)
Chunking	split by article and by numbered paragraph, about 900 characters per chunk
Result	**559 chunks** (294 from articles, 265 from recitals), median 673 characters

Each chunk starts with its article title, so it keeps its topic. Example chunk (`Art. 33(1-2)`):

> *GDPR Article 33 – Notification of a personal data breach to the supervisory authority*
> 1. In the case of a personal data breach, the controller shall without undue delay and, where feasible, not later than 72 hours after having become aware of it, notify the personal data breach to the supervisory authority [...], unless the personal data breach is unlikely to result in a risk to the rights and freedoms of natural persons. [...]
> 2. The processor shall notify the controller without undue delay after becoming aware of a personal data breach.

2 Models and setup

Role	Model / method	Details
Retrieval 1 (dense)	`all-MiniLM-L6-v2` (Bi-Encoder)	22.7M parameters, 6 layers, 384-number vectors, max 256 tokens; stored in a FAISS `IndexFlatIP` (exact cosine search)
Retrieval 2 (sparse)	BM25 (Okapi, `rank-bm25`)	keyword scoring, no neural model; lower-case word split; $k_1 = 1.5$, $b = 0.75$
Re-ranker	`ms-marco-MiniLM-L-6-v2` (Cross-Encoder)	22.7M parameters, 6 layers, max 512 tokens; trained to rank passages for search queries (MS MARCO)
LLM (generator)	`Qwen2.5-3B-Instruct`	3.09B parameters, 36 layers, 32k-token context; open weights (Hugging Face), run locally; greedy decoding, max 200 new tokens
Hardware	laptop CPU	Intel Core i5-1245U, 30 GB RAM, no GPU
Software	Python 3.14	PyTorch 2.14 (CPU), transformers 5.19, sentence-transformers 6.1, faiss-cpu 1.15, rank-bm25 0.2.2

Which "thinking" model? None. `Qwen2.5-3B-Instruct` is a normal *instruction-following* model, not a reasoning ("thinking") model: it writes the answer directly, without a hidden step-by-step reasoning

phase. We chose it because the assignment allows local open-weights models, no API key was needed, it runs on a laptop CPU, and in RAG the answer should come from the retrieved chunks rather than from the model's own reasoning.

3 Pipeline approaches

All three pipelines use the same 559 chunks, the same LLM and the same prompt.

Step	Pipeline 1 – Dense baseline	Pipeline 2 – Dense + re-ranking	Pipeline 3 – BM25
1. Index	chunks embedded once by the Bi-Encoder, stored in FAISS	the same index	BM25 keyword index of the chunks
2. Retrieve	Top-10 by cosine similarity	the same Top-10 as Pipeline 1	Top-10 by BM25 score
3. Re-rank	–	Cross-Encoder scores the 10 (question, chunk) pairs; sort by score	–
4. Context	Top-3	new Top-3	Top-3
5. Generate	LLM → **Answer 1**	LLM → **Answer 2**	LLM → **Answer 3**

Prompt (same for all). *System:* "You are a careful legal assistant answering questions about the EU General Data Protection Regulation (GDPR). Answer ONLY from the numbered context passages. State the exact figures, conditions and exceptions they contain and cite the passage reference in brackets, e.g. [Art. 33(1)]. If the passages do not contain the answer, reply: 'The provided context does not answer this question.' Keep the answer under 120 words." *User:* the 3 context chunks, each with its reference, then the question.

4 Architecture comparison

Bi-Encoder (dense retrieval): the question and each chunk are turned into vectors *separately*, and we compare the vectors with cosine similarity. Chunk vectors are made once in advance, so search is very fast, and it can match words with similar meaning. But a whole chunk is squeezed into one vector, so small words like *not* or *unless* are easily lost.

BM25 (sparse / keyword retrieval): no neural model. A chunk scores high if it contains the question's exact words, especially rare ones. It is the fastest method and works well when the question uses the same words as the document, but it cannot match synonyms ("fired" vs "dismissed") or different word forms ("objects" vs "object").

Cross-Encoder (re-ranking): the question and one chunk go into the model *together*, so every word of the question can look at every word of the chunk (self-attention). It judges relevance much better, but it must run once for every (question, chunk) pair, so it is slow and is only used on the 10 candidates.

5 Example output (normal question)

When the question uses the same words as the law, all three pipelines find the right chunk first and give the same correct answer.

Question: *Within how many hours must a personal data breach be reported to the supervisory authority?*

Rank	Bi-Encoder	+ Cross-Encoder	BM25
1	Art. 33(1-2)	Art. 33(1-2)	Art. 33(1-2)
2	Recital 85	Recital 85	Recital 85
3	Recital 87	Art. 33(5)	Art. 33(3-4)

Answer 1 (Bi-Encoder)	A personal data breach must be reported to the supervisory authority within 72 hours after becoming aware of it [Art. 33(1)].
Answer 2 (re-ranked)	A personal data breach must be reported to the supervisory authority within 72 hours after becoming aware of it [Art. 33(1-2)].
Answer 3 (BM25)	A personal data breach must be reported to the supervisory authority within 72 hours after becoming aware of it [Art. 33(1)].
Result	All three correct (Art. 33(1): "not later than 72 hours"). Differences only appear when the right chunk is *not* already in the Top-3, as in the failures below.

6 Failure outputs: qualitative and rank-shift analysis

For each question, the "gold" chunk is the chunk that contains the answer in the law. Only ranks 1–3 are given to the LLM.

ID	Question	Gold chunk	Bi-Encoder	+ Cross-Encoder	BM25
Q1	What is the highest penalty for a company that violates people's rights under GDPR?	Art. 83(5)	#4	#1	not in Top-10
Q2	Do I have to notify affected users of a breach if the stolen data was encrypted?	Art. 34(3)	#4	#1	#3
Q3	Can the data protection officer be fired for doing their job?	Art. 38(3)	#5	#1	#5
Q4	Does a company have to stop processing if a person objects?	Art. 21(1)	#9	#1	not in Top-10
Q5	How quickly must a company fix wrong information about me?	Art. 16	#9	#3	not in Top-10

Example (Q1). The word "penalty" made the Bi-Encoder put Article 84 (titled "Penalties") first, but the real answer, Art. 83(5), was only 4th. The Cross-Encoder moved it to 1st. BM25 did not find it at all:

Rank	Bi-Encoder	+ Cross-Encoder	BM25
1	Art. 84 "Penalties" (wrong)	**Art. 83(5) – gold**	Recital 56
2	Art. 83(4) (lower fine tier)	Art. 83(4)	Art. 18(2-3)
3	Recital 150	Recital 150	Recital 151
4	**Art. 83(5) – gold**	Art. 84 "Penalties"	(gold not in Top-10)

Answer 1 vs. Answer 2 vs. Answer 3 (exact LLM answers):

Q1 – Keyword trap	gold Art. 83(5): Bi-Encoder #4, Cross-Encoder #1, BM25 not in Top-10
Question	What is the highest penalty for a company that violates people's rights under GDPR?
Answer 1 (Bi-Encoder)	The highest penalty for a company that violates people's rights under GDPR is 2% of the company's global annual turnover of the previous financial year, whichever is higher. [Art. 83(4)]
Answer 2 (re-ranked)	The highest penalty for a company violating people's rights under GDPR is 4% of the company's global annual turnover, or up to 20,000,000 EUR, whichever is greater. [Art. 83(5)]
Answer 3 (BM25)	The provided context does not answer this question.
Result	Answer 1 **misstates the rule** (the lower 2% tier, Art. 83(4)). Answer 2 is **correct**: EUR 20 million or 4%, whichever is higher. Answer 3 **cannot answer**.
Why	The question says "penalty", and Art. 84 is titled "Penalties", so the Bi-Encoder ranked it 1st, even though it only lets countries add their own penalties; Art. 83(4), a lower fine, also looked similar. The right chunk, Art. 83(5), uses different words ("administrative fines", "data subjects' rights"), so it was only 4th. The Cross-Encoder reads the question and the chunk together, so it matched "people's rights" with "data subjects' rights" and put Art. 83(5) 1st. BM25 only matches exact words, and Art. 83(5) contains none of "penalty", "highest" or "company", so it was not even in BM25's Top-10.

Q2 – Exception	gold Art. 34(3): Bi-Encoder #4, Cross-Encoder #1, BM25 #3
Question	Do I have to notify affected users of a breach if the stolen data was encrypted?
Answer 1 (Bi-Encoder)	The provided context does not directly state whether encryption of data affects the notification obligation. However, it mentions that if the controller can demonstrate that the personal data breach is unlikely to result in a risk to the rights and freedoms of natural persons, they are not required to notify the supervisory authority. [Art. 33(2)]
Answer 2 (re-ranked)	Yes, according to Art. 34(3)(a), if the controller has implemented appropriate technical and organizational measures, such as encryption, and those measures were applied to the personal data affected by the breach, then notification to the data subjects is not required.
Answer 3 (BM25)	Yes, you have to notify affected users if the stolen data was encrypted [Art. 34(3)(a)]. This is because encryption is mentioned as a measure that renders personal data unintelligible to unauthorized individuals, thus meeting the condition for not requiring notification to users.
Result	Answer 1 **misses the exception**. Answer 2 is **correct**: with encryption, telling the users is not required. Answer 3 is **wrong and contradicts itself** ("Yes, you have to notify", then the opposite).
Why	The Bi-Encoder's Top-3 were the general data-breach rules (Art. 33 and Recitals 85 and 87), which share many words with the question. The exception (Art. 34(3)) is about the same topic, and the deciding word "encryption" is only one word in a long chunk, so it ended up 4th. The Cross-Encoder saw "encrypted" in the question next to "such as encryption" and ranked the exception 1st. BM25 found it 3rd (shared words "breach", "data"), but behind two general breach rules, and the LLM followed the general rule.

Q3 – Negation / role limit	gold Art. 38(3): Bi-Encoder #5, Cross-Encoder #1, BM25 #5
Question	Can the data protection officer be fired for doing their job?
Answer 1 (Bi-Encoder)	According to Art. 38(5), the data protection officer is bound by secrecy concerning the performance of their tasks, in accordance with Union or Member State law. This implies that the officer cannot be fired for doing their job as it would involve breaching confidentiality. [Art. 38(5)]
Answer 2 (re-ranked)	Yes, the data protection officer cannot be dismissed or penalized by the controller or processor for performing their tasks [Art. 38(3)].
Answer 3 (BM25)	No, the data protection officer cannot be fired for doing their job according to [Art. 38(6)]. This passage states that the controller or processor shall ensure that any tasks and duties of the data protection officer do not result in a conflict of interests. Therefore, the officer's role is protected from being terminated due to the performance of their duties.
Result	Answer 1 **makes up a reason** (secrecy, Art. 38(5)). Answer 2 is **correct**: the DPO cannot be dismissed for doing their job. Answer 3 also **makes up a reason** (conflict of interest, Art. 38(6)).
Why	All the chunks about the data protection officer (Art. 37–39) look alike to the Bi-Encoder, and the law says "shall *not* be dismissed", not "fired"; the word "not" barely changes a vector. BM25 had the same problem: it can only match "data protection officer", which is in every DPO article, so it also ranked the right chunk 5th. The Cross-Encoder linked "fired" to "dismissed" and ranked Art. 38(3) 1st.

Q4 – Buried at rank 4–10	gold Art. 21(1): Bi-Encoder #9, Cross-Encoder #1, BM25 not in Top-10
Question	Does a company have to stop processing if a person objects?
Answer 1 (Bi-Encoder)	The provided context does not directly state whether a company has to stop processing if a person objects. However, GDPR Article 6(1)(e) states that processing is necessary for the performance of a task carried out in the public interest or in the exercise of official authority vested in the controller. This implies that if an individual objects, the company might need to reconsider the necessity of processing their data for tasks related to public interest or official authority, but it does not explicitly mandate stopping processing. [Art. 6(1)(e)]
Answer 2 (re-ranked)	Yes, if a person objects, the company has to stop processing the personal data unless there are compelling legitimate grounds for the processing that override the interests, rights and freedoms of the data subject [Art. 21(1-2)].
Answer 3 (BM25)	Yes, a company must stop processing if a person objects, as stated in Art. 17(1)(c) [Art. 17(1)] where it says "the data subject objects to the processing pursuant to Article 21(1) and there are no overriding legitimate grounds for the processing".
Result	Answer 1 **misses the rule** and cites the wrong article. Answer 2 is **correct**: stop, unless there are compelling legitimate grounds. Answer 3 is **partly correct**: right idea, but taken from the right-to-erasure article (Art. 17(1)).
Why	The question uses everyday words ("company", "stop", "objects"), while the law uses legal words ("controller", "shall no longer process", "object"). The Bi-Encoder ranked general chunks about processing first and the right chunk 9th. BM25 matches whole words exactly ("objects" $\neq$ "object"), so it missed Art. 21(1) completely. The Cross-Encoder matched "stop processing" with "no longer process" and ranked it 1st.

Q5 – Keyword trap (synonym)	gold Art. 16: Bi-Encoder #9, Cross-Encoder #3, BM25 not in Top-10
Question	How quickly must a company fix wrong information about me?
Answer 1 (Bi-Encoder)	The context does not directly address how quickly a company must fix wrong information about you. It discusses notifying authorities about personal data breaches but does not specify a timeframe for correcting inaccurate information. [Art. 33(1-2)]
Answer 2 (re-ranked)	According to GDPR Article 16(1), a company must fix wrong information about you without undue delay. However, there is no specific timeframe mentioned for how quickly this must be done. [Art. 16(1)]
Answer 3 (BM25)	The provided context does not answer this question.
Result	Answers 1 and 3 **cannot answer**. Answer 2 is **correct**: wrong data must be corrected "without undue delay".
Why	"How quickly" matched the time words in the data-breach rules ("without undue delay", "72 hours"), so two of the Bi-Encoder's Top-3 were breach rules. The real answer is the "right to rectification" (Art. 16), which never uses "fix" or "wrong", so it was 9th for the Bi-Encoder and not in BM25's Top-10 at all. The Cross-Encoder understood that "fix wrong information" means "rectification of inaccurate data" and moved Art. 16 to 3rd.

Summary: only Pipeline 2 (Cross-Encoder re-ranking) put the gold chunk in the Top-3 for all five questions, and all five Answers 2 were correct. The Bi-Encoder never had it in the Top-3, and BM25 had it there only once (Q2) and missed it completely for Q1, Q4 and Q5. So Answers 1 and 3 misstated rules, missed exceptions, made up reasons, or could not answer.

Why does this happen?

- **Bi-Encoder:** it turns each chunk into one vector *before* it sees the question, so it matches the *topic*: a chunk with the same word but a different meaning can win (Q1, Q5), and small words like "not" or one key word like "encryption" hardly change a long chunk's vector (Q2, Q3).
- **BM25:** it needs the *exact* same words. Questions in everyday words ("penalty", "fix", "stop", "objects") do not match the law's words ("fines", "rectification", "no longer process", "object"), so the right chunk is not found (Q1, Q4, Q5).
- **Cross-Encoder:** it reads the question and the chunk *together*, word by word, so it can link synonyms and see the specific condition. Because the LLM only reads the Top-3 chunks, this re-ordering decides whether the LLM ever sees the right rule.

7 Latency profiling

Measured on a laptop CPU (average of 10 runs per question for retrieval; LLM time is the average over the five questions).

Step	**Pipeline 1 Bi-Encoder**	**Pipeline 2 + Cross-Encoder**	**Pipeline 3 BM25**
Bi-Encoder retrieval (Top-10)	12.3 ms	12.3 ms	–
BM25 retrieval (Top-10)	–	–	2.5 ms
Cross-Encoder re-ranking (10 pairs)	–	294.0 ms	–
Total retrieval	**12.3 ms**	**306.3 ms**	**2.5 ms**
LLM answer	58.5 s	55.4 s	55.3 s
End-to-end	**58.6 s**	**55.7 s**	**55.3 s**

BM25 is the fastest retriever and the Bi-Encoder is also fast. Re-ranking makes retrieval about 25× slower than the Bi-Encoder alone, but it adds only 0.29 s, which is under 1% of the total time, because the LLM takes much longer.

8 Scaling question

The Cross-Encoder cannot pre-compute anything: it must read the question together with *every* chunk, again for every new question. At about 29 ms per chunk, all 559 chunks would take about 16 s per question (about 8 hours for 1 million chunks). The Bi-Encoder (about 12 ms) and BM25 (about 2 ms) search all chunks quickly because their indexes are built once in advance. So we use both: a fast retriever picks 10 candidates, and the accurate Cross-Encoder re-ranks only those 10.